



Figure 1: WEKA screenshot

component-based Knowledge Flow interface as well as from the command line. There is also a module called *Experimenter*, which allows the systematic comparison of the predictive performance of machine learning algorithms on a collection of datasets. *Explorer* has several panels providing access to the main components of the workbench: the *Preprocess* panel has facilities for importing data from a database, a CSV file, etc, and to preprocess this data using a filtering algorithm. Such filters can be used to transform the data and make it possible to delete instances and attributes as per specific criteria. The *Classify* panel provides the features to apply classification and regression algorithms to the dataset, to estimate the accuracy of the resulting predictive model and visualise erroneous predictions, ROC curves or the model. The *Associate* panel provides the access for association rule learning to identify the interrelationships between attributes in the data. The *Cluster* panel or module provides access to the clustering techniques, including simple k-means algorithm and many others. The *Select attributes* panel provides access to the algorithms for the identification of the most predictive attributes in a dataset. The *Visualize* panel depicts a scatter plot matrix in which individual scatter plots can be selected, enlarged and analysed using various selection operators.

Data file formats in WEKA

WEKA can execute the data files in multiple formats including ARFF, CSV, XML, LIBSVM, BSI, DAT and C4.5. A spreadsheet can be easily converted to CSV (Comma Separated Values). WEKA imports the CSV file format and any machine learning algorithm can be applied.

There are a number of file formats, but classically, the ARFF (Attribute Relationship File Format) is widely used for analysis of datasets.

ARFF files have two distinct sections. The first is the header information, which is followed by the data

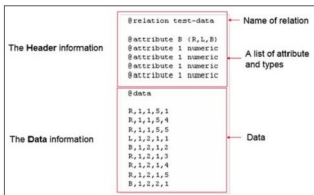


Figure 2: ARFF file

information. Lines that begin with a % are comments. The *@RELATION*, *@ATTRIBUTE* and *@DATA* declarations are case insensitive. The *ARFF Header* section of the file contains the relation declaration and attributes declarations. Each attribute in the data set has its own *@* attribute statement, which uniquely defines the name of that attribute and its data type.

The ARFF file can be easily viewed in the table format by the ARFF viewer of the WEKA tool. Using this option, the typical ARFF file can be viewed in tabular form.

One exceptional feature of WEKA is the database connection using JDBC with any RDBMS package. Let's suppose that you have collected data from a government organisation or agency. If the data collected is in database format—MySQL, MS Access, DB2 or any other, then there is no need to worry about whether we will have to convert the millions of datasets in the WEKA-specific file formats. We can simply connect WEKA with any database management software using JDBC, and can import any number of datasets of that database server.

Implementation of the Apriori algorithm

The Apriori algorithm is the most widely used algorithmic approach for frequent item set mining as well as association rule learning on transactional databases. It works by identifying the frequent individual items existing in the database, and enhancing them to larger and larger item sets as long as those item sets show up often in the database. The frequent items extracted by Apriori can be used to establish the association rules, which emphasise general trends in the database. This has applications in domains such as market basket analysis, forensics, agriculture and many others.

Implementation of the OneR/1R/1-Rule classification

The 1R learning algorithm is the rule-based classification learning algorithm for discrete attributes. The algorithm